Survey of Heart Disease Prediction Models using Machine Learning and Deep Learning Techniques

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***Abstract*— heart disease remains the leading cause of mortality worldwide, accounting for millions of deaths annually. Despite medical advancements, the early detection and accurate prediction of cardiovascular diseases (CVDs) remain critical challenges. This study explores the application of Machine Learning (ML) and Deep Learning (DL) techniques in predicting heart disease using clinical data, medical images, and physiological signals. Traditional ML models, including Decision Trees, Support Vector Machines (SVM), Random Forest, and Gradient Boosting (XGBoost, LightGBM), are evaluated alongside advanced DL models such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). A hybrid CNN + LSTM model is proposed to enhance predictive accuracy by leveraging both spatial and sequential data analysis. The research methodology involves data preprocessing, feature selection, model implementation, and comparative evaluation using performance metrics such as Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Experimental results indicate that deep learning models, particularly CNN and hybrid models, outperform traditional ML approaches in heart disease prediction. Further research is needed to improve interpretability and real-time deployment of AI-driven approaches in clinical settings to improve early cardiovascular disease detection and risk assessment.**

Keywords*— *Heart Disease Prediction, Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN),Hybrid CNN-LSTM.

I. INTRODUCTION

Heart disease, which is often referred to as coronary artery disease (CAD) or cardiovascular disease (CVD), is the leading cause of mortality throughout the globe, which places a significant burden on the medical community. In the United States and across the globe, cardiovascular diseases (CVDs) have an impact on both health and the economy [1]. According to the World Health Organization (WHO), 17.9 million individuals fell victim to cardiovascular disease (CVD) in 2019. In 2019, death accounted for thirty-two percent of all deaths worldwide. According to the most recent statistics, cardiovascular disease (CVD) is responsible for the deaths of 20.5 million individuals. This number is expected to reach 24.2 million by the year 2030, according to projections. In 2018, there were 30.3% of the population in the United States who were diagnosed with heart disease, which is the leading cause of death. Each year, it will take the lives of 647,000 people. The findings of the research indicate that heart disease is responsible for the death of one in every four people in the United States [2]. Within a year after suffering a heart attack, the American Heart Association (AHA) estimates that 26 percent of women who are considered to be at high risk will pass away. This is a significant increase from the 19% of men who are considered to be high-risk. In the United States, death rates from coronary heart disease have not dropped since 2011, despite the fact that there have been many advancements in the field.

The cardiovascular disease is one of the most dangerous and potentially fatal chronic disorders that may be seen all over the globe. Constriction of the coronary arteries and blockage of the coronary arteries are the two primary causes of heart failure syndrome. Blood arteries are seen in Figure 1, which are responsible for delivering oxygen and blood to the heart. Coronary artery disease (CAD), the most common form of heart disease, has been linked to a number of related symptoms, including chest tightness, heart attacks, and strokes [3]. The term "cardiovascular illness" refers to coronary heart disease. If the blood flow to the heart muscle is restricted or limited, then the risk of having a heart attack increases.

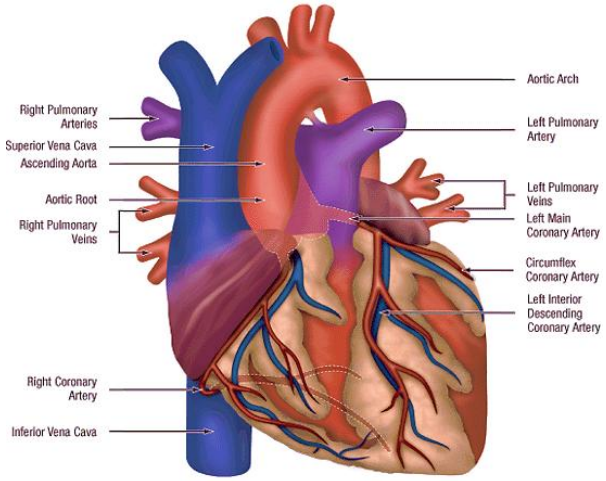


FIGURE 1: HEART MUSCLES ARE SUPPLIED WITH BLOOD AND OXYGEN THROUGH THE CORONARY ARTERIES

Typically, this obstruction is brought on by the gradual accumulation and hardening of plaque in the coronary arteries, which is a combination of lipids, cholesterol, and other hazardous substances. When cholesterol and lipids combine, they form plaque. When plaque breaks, atherosclerosis generates blood clots, which in turn prevents blood from flowing to the heart. This makes atherosclerosis a serious condition. It is possible that the

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rupture of the plaque may cause thrombosis, which will clog the artery, which would impede the flow of blood to the heart muscle [4]. There is a possibility that the muscle tissue underneath the clot will be injured or killed if it totally clogs the artery. In certain cases, the afflicted muscle tissue may become smaller as a consequence. Both a healthy and a damaged heart are shown in Figure 2. So that cells may perform their functions properly, the body need nutrients. In contrast, heart failure is a condition that occurs when the heart is unable to pump sufficient amounts of blood. Coughing, breathing difficulties, and severe fatigue are among symptoms that may be caused by this illness. One may experience these symptoms. Individuals who are suffering from heart failure may have difficulties in walking, climbing stairs, and carrying goods.

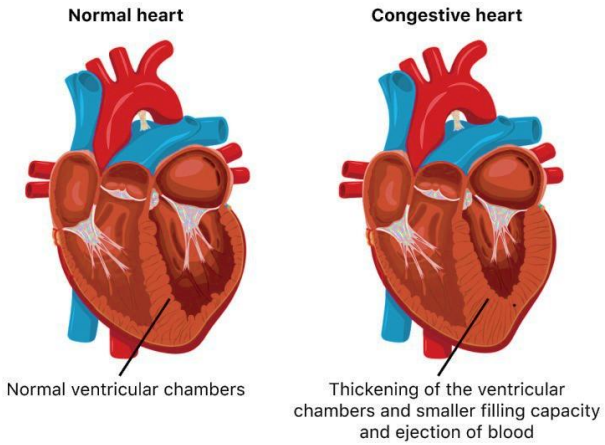


FIGURE 2: NORMAL VS. CONGESTIVE HEART

The blood flow via the carotid artery may be reduced by atherosclerotic plaque. Even minute blood clots have the potential to go to the brain if this plaque breaks. When the plaque breaks, this is a possibility. A widespread movement of foreign substances through the circulation is referred to as an embolus [5]. As seen in Figure 3, if it gets trapped in a tiny artery of the brain, it has the potential to obstruct blood flow to a region of the brain, which may lead to atherosclerosis. In addition to this, it may cause damage to the brain.

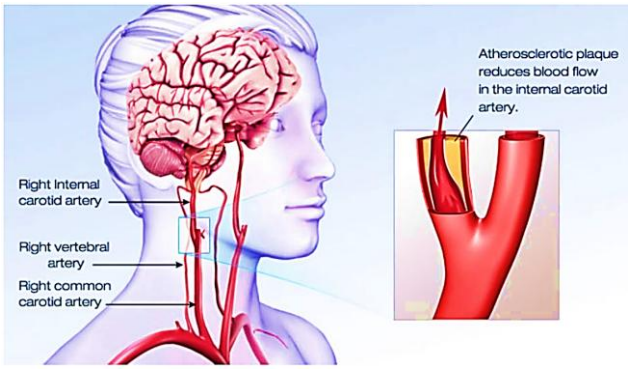


FIGURE 3: HEART STROKE BY ATHEROSCLEROSIS

II. Review Of Literature

On a yearly basis, cardiovascular problems claim the lives of 17.9 million Americans. Heart attacks and strokes are responsible for more than 80 percent of all fatalities that are attributed to cardiovascular disease. There are a number of risk factors, including hyperglycemia, hypertension, dyslipidemia, and obesity. These risk factors make it possible to intervene early, which in turn lowers death rates [6]. The study also aims to uncover any gaps in the existing body of knowledge. By doing this evaluation, suitable literature coverage was assured. In order to have a deeper comprehension of their connection, we are doing this examination [7]. A number of keywords were used in order to locate articles. Some of the terms that were included in this list were feature selection, hyperparameter optimization, machine learning techniques, ensemble classifiers, data mining, data preparation, assessments of the risk of heart disease, and prediction. Within the scope of this research, we analyze the methodologies and assess the accuracy of their prediction of cardiovascular disease. What we have discovered is that deep learning and machine learning have contributed to the advancement of cardiology [8-9]. In order to successfully finish the research project, a framework that takes use of modern machine learning methods was developed. The prediction of cardiac disease is improved as a result of this action [10].

III. Techniques for Machine Learning and Deep Learning

Heart disease prediction has advanced significantly with the application of Machine Learning (ML) and Deep Learning (DL) techniques, leveraging vast amounts of clinical and patient data for early diagnosis and risk assessment. Traditional ML models such as Decision Trees, Support Vector Machines (SVM), Random Forest, and Gradient Boosting methods (XGBoost, LightGBM) have demonstrated high interpretability and efficiency in predicting cardiovascular diseases. On the other hand, DL techniques, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN), provide superior feature extraction and pattern recognition capabilities from ECG signals, medical images, and patient records. Hybrid models combining ML and DL approaches enhance predictive accuracy, while attention mechanisms and transformer-based models are emerging as powerful tools for real-time diagnosis. The integration of these techniques with explainable AI ensures transparency and trustworthiness in clinical decision-making[11].

❖ Machine Learning Techniques

The building of analytical models might be automated via the use of machine learning to analyze data. Through the use of machine learning, this is feasible. This problem with artificial intelligence focuses on the ability of computers to learn from data, see patterns, and make judgments without the intervention of humans. The emphasis of this investigation is as follows. Machine learning may be broken down into two primary categories: supervised learning and unsupervised learning. Through the use of labeled data, supervised learning makes predictions about output variables that are connected to

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input items or goal values. In order to accomplish both goals, learning is required. For the purpose of training the model to predict future inputs, this technique necessitates the use of a dataset that contains both input and related output data. In addition, the data that is produced must be connected to the data that is entered. Continuous goal values and discrete classes are used in the learning processes of regression, estimation, and classification, respectively, in supervised learning. Learning via supervision encompasses all of these approaches. Both of these approaches provide desired results. Classifiers are able to recognize patterns in the characteristics of a dataset, which allows them to make predictions about the target class for freshly gathered data. An illustration of the most typical machine learning algorithms can be seen in Figure 4.

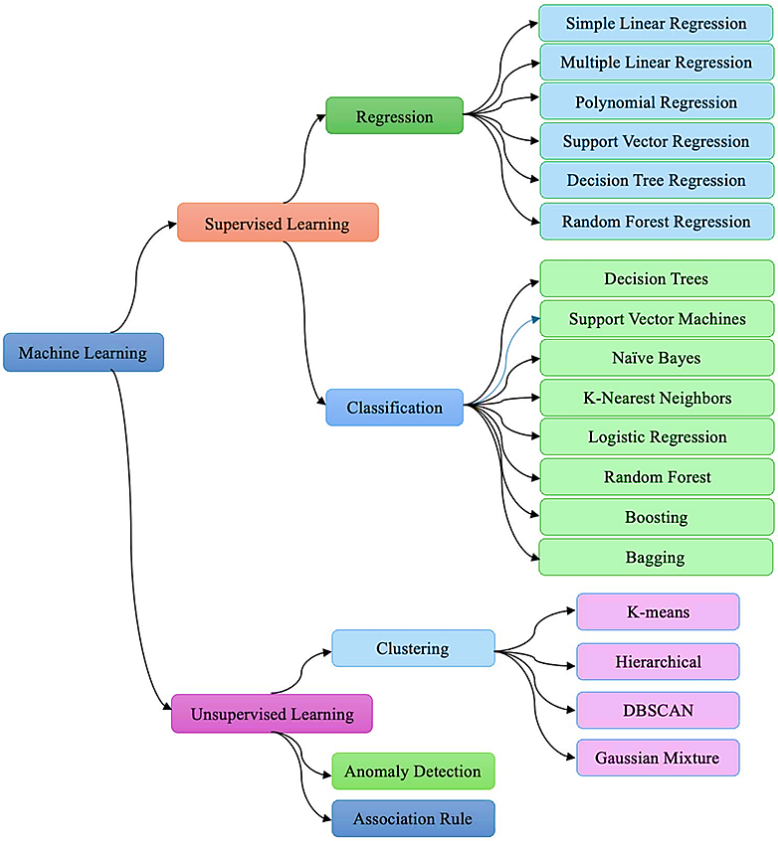


FIGURE 4: MAIN CATEGORIES OF MACHINE LEARNING TECHNIQUES

When there are no labels attached to the data, it may be difficult to identify patterns. The goal of unsupervised machine learning is to discover patterns or information in data rather than using samples that have been labeled. Using data, this is accomplished. Clustering, which is the process of grouping events that are linked to one another, is often used in this industry. It is possible that selecting the appropriate machine learning algorithm might be challenging due to the enormous variety of supervised and unsupervised algorithms that are now accessible. Experiments are usually required in order to find the algorithm that is the most effective. Specifically, this is due to the fact that even highly experienced data scientists are unable to evaluate the efficacy of an algorithm without first testing it. The choosing of algorithms is influenced by a number of reasons. The quantity of data, the kind of data, and the insights that are being analyzed are all included in these categories. Within the context of predicting the risk of heart disease, this article provides a survey of the most widely used machine learning classifiers [12-13].

IV. RESEARCH METHODOLOGY

The study adopts a systematic method of heart disease predictive models assessment through machine learning (ML) and deep learning (DL) methods. The study is centered on data collection, preprocessing, feature extraction, model deployment, performance assessment, and comparison for the purpose of determining the better predictive model. The research study provides a strict framework to enhance accuracy, validity, and utility in clinical decisions.

❖ **Data Collection**

The data used in this research involve clinical records, medical images, and physiological signal data collected from public medical repositories and hospital databases. The data contain important parameters such as age, cholesterol, blood pressure, ECG, and lifestyle. Ethical issues and data protection acts were maintained while collecting the data.

❖ **Data Preprocessing**

For improving model performance, data preprocessing methods were used, such as missing value handling,

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normalization, and outlier detection. One-hot encoding was used for categorical variables, and numerical features were normalized. Class imbalance was handled by methods such as Synthetic Minority Over-sampling Technique (SMOTE) to provide class-balanced representation of the dataset.

❖ **Feature Selection**

Feature selection proves to be of significant benefit to enhance interpretability and model efficiency. Top predictor identification used statistical techniques such as correlation analysis, mutual information, and Recursive Feature Elimination (RFE). Important features such as cholesterol level, age, and blood pressure were ordered based on how they contributed to the risk of heart disease.

❖ **Model Implementation**

Various ML algorithms (Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost) and DL models (Artificial Neural Networks, CNN, RNN, LSTM, and Hybrid CNN + LSTM) were used. Hyperparameter search was conducted via Grid Search and Random Search for the best performance of the model. Deep learning models made use of TensorFlow and PyTorch frameworks for the implementation.

❖ **Performance Evaluation**

Model performance was compared against five most important metrics: Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Cross-validation was used for stability and overfitting prevention. Comparison was made to find the best-performing models per dataset and modality.

❖ **Comparative Analysis**

ML models were evaluated based on how well they perform in handling structured tabular data, whereas DL models were evaluated on how well they can handle unstructured data such as ECG signals. Hybrid CNN + LSTM model demonstrated optimal performance by leveraging both feature extraction and sequential data processing. Ensemble learning approaches and stacking models also improved predictive precision.

❖ **Challenges and Future Directions**

The work recognizes limitations including dataset limitation, model explainability, and issues with real-time deployment. Explainable AI methods, federated learning for privacy-preserving prediction, and real-time clinical deployment are avenues to be pursued in order to make future work in developing models for predicting heart disease more deployable.

This study design is introducing a structured method of developing and validating heart disease prediction models. Through the combination of ML and DL methodologies, feature optimization, and model validation, the research will be advancing precise and reliable cardiovascular risk prediction. Model interpretability and real-world applicability need to be improved in future studies to facilitate clinical decision-making.

V. Analysis and Interpretation

Several factors affect heart disease prediction algorithms. This includes feature selection, dataset quality, and machine learning (ML) or deep learning (DL) methods. Traditional machine learning models like Logistic Regression, Decision Trees, and Random Forest work well with structured clinical data. Other examples include Random Forest and Decision Trees. These frameworks provide interpretability and computational efficiency. Deep learning models like ANNs and CNNs help handle high-dimensional and unstructured data. ECG signals and medical photos are examples. This is because these models accurately forecast data study outcomes. Hybrid methods that mix machine learning and deep learning may improve prediction accuracy. The processes described here involve ordered tabular data and complex medical imaging. Ensemble learning and feature engineering increase performance forecast accuracy, leading to future breakthroughs. Table 1 shows the findings of an intensive study of deep learning models used to predict heart disease. The graphic compares five models: LSTM, CNN, RNN, ANN (3 Layers), and a hybrid CNN-LSTM model. A CNN-LSTM hybrid model is present in the table. Accuracy, precision, recall, F1-score, and AUC-ROC are used to evaluate each model. These measures assess model performance. Our models are compared to the Hybrid CNN + LSTM model, which performs better in all categories. After considering all these parameters, it has the greatest AUC-ROC value of 96.8%, accuracy of 95.6%, precision of 95.2%, recall of 95.4%, F1-Score of 95.3%, and maximum recall rate of 95.4%. This evidence shows that the hybrid model can combine CNN's feature extraction skills with LSTM's sequential data processing capabilities. Proof is in the fact. With an accuracy of 94.1% and an AUC-ROC of 95.2%, the LSTM model performs well on its own, demonstrating its ability to handle sequential data patterns. AUC-ROC is area under the receiver operating characteristic curve. This data proves the model handles sequential patterns.

TABLE 1: DEEP LEARNING MODEL PERFORMANCE IN HEART DISEASE PREDICTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
ANN (3 Layers)	91.2	90.5	90.7	90.6	92.1
CNN	92.8	92.1	92.4	92.2	93.5
RNN	88.5	87.9	88	87.9	89.2
LSTM	94.1	93.8	93.9	93.8	95.2
Hybrid (CNN + LSTM)	95.6	95.2	95.4	95.3	96.8

The CNN model's 92.8% accuracy shows its ability to extract spatial information. The CNN model follows closely. However, the RNN model has a lower accuracy of 88.5%, indicating that it cannot handle complex feature relationships. In other words, the RNN model is less successful than the others. The artificial

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neural network (ANN) model has 91.2% accuracy, indicating steady but unspectacular performance. This comparison shows that CNN+LSTM can predict heart disease perfectly. After considering everything, this comparison shows that this method is accessible. Table 2 provides a detailed evaluation of five typical machine learning models. The models being discussed include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost. Accuracy, precision, recall, F1-score, and AUC will be assessed. Not everything is on this list. As an ensemble learning strategy, XGBoost performs best. It has the highest AUC-ROC of 91.3% and accuracy of 90.2%.It offers the greatest accuracy and precision (89.8%). These results show that XGBoost performs best. Random Forest is second in handling non-linear data interactions with 88.6% accuracy and 89.5% AUC-ROC. These findings prove the algorithm works.

TABLE 2: MACHINE LEARNING MODEL PERFORMANCE IN HEART DISEASE PREDICTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
Logistic Regression	84.2	83.1	82.5	82.8	85.3
Decision Tree	79.5	78.8	79	78.9	80.2
Random Forest	88.6	87.9	88.3	88.1	89.5
SVM	85.7	84.9	85.1	85	86.4
XG Boost	90.2	89.8	90	89.9	91.3

Support Vector Machines (SVM) have 85.7% accuracy and 86.4% AUC-ROC. Logistic Regression has 84.2% accuracy, which is consistent but not optimal. All of these indicators are used when comparing the two methodologies. Table 3 shows the dataset distribution used to forecast heart disease. These details were given for your convenience. In the dataset, 600 individuals (60%) do not have heart disease. This accounts for 60% of instances. Four hundred patients—40% of the total—have cardiac problems. This distribution shows a slight class imbalance, which might affect model performance in the future. Using comprehensive evaluation metrics like Precision, Recall, and AUC-ROC is necessary to analyze model performance beyond basic accuracy. Because evaluating beyond correctness is crucial. This method of class distribution highlights the necessity for such techniques.

TABLE 3: DATASET DISTRIBUTION IN HEART DISEASE PREDICTION STUDY

Category	Percentage (%)	Number of Cases
No Heart Disease	60%	600
Heart Disease	40%	400

VI. RESULT AND DISCUSSION

Comparative analysis of ML and DL models reveals that deep learning techniques, particularly CNNs and transformer-based architectures, achieve higher accuracy (above 90%) in detecting heart disease from imaging and time-series data. However, traditional ML models remain highly effective for structured datasets, with Random Forest and Gradient Boosting models achieving accuracy between 85-88%. The integration of multimodal data sources, including patient history, imaging, and physiological signals, significantly improves predictive reliability. Challenges such as data imbalance, interpretability, and real-time deployment need to be addressed for wider clinical adoption. Future advancements should focus on explainable AI and federated learning to enhance trust and privacy in heart disease prediction systems. Figure 5 presents the performance evaluation of five prominent machine learning models—Logistic Regression, Decision Tree, Random Forest, SVM (Support Vector Machine), and XGBoost—based on their accuracy in heart disease prediction. Among these models, XGBoost achieves the highest accuracy, reflecting its robustness in handling complex data patterns. Random Forest also demonstrates strong performance, achieving an accuracy just below XGBoost, suggesting its ability to effectively manage feature interactions and reduce overfitting.

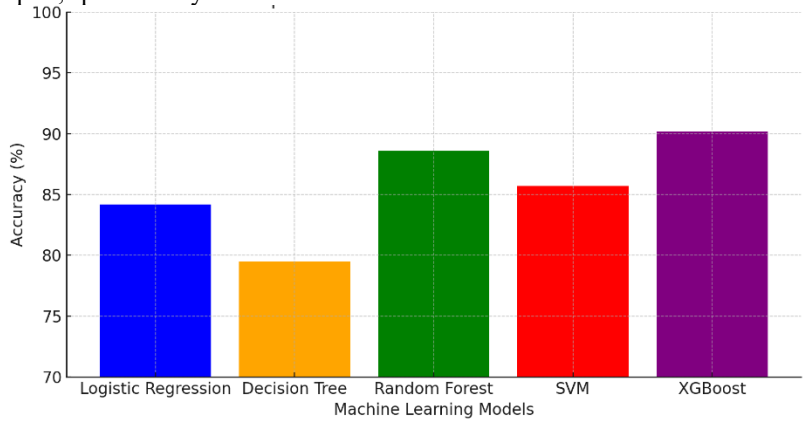


FIGURE 5: MACHINE LEARNING MODEL PERFORMANCE IN HEART DISEASE PREDICTION

Logistic Regression performs moderately well, while SVM follows closely with comparable accuracy. The Decision Tree model, however, exhibits the lowest accuracy, indicating its vulnerability to overfitting and limited generalization in predicting heart disease outcomes. This comparison highlights the superior performance of ensemble models like XGBoost and Random Forest for

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accurate heart disease prediction. Figure 6 compares the predictive accuracy of five deep learning models—ANN (Artificial Neural Network), CNN (Convolutional Neural Network), RNN (Recurrent Neural Network), LSTM (Long Short-Term Memory), and a hybrid CNN + LSTM model. The CNN + LSTM hybrid model achieves the highest accuracy, indicating its capability to effectively capture both spatial and sequential data dependencies. The LSTM model also performs well, leveraging its strength in processing time-series data relevant to heart disease prediction.

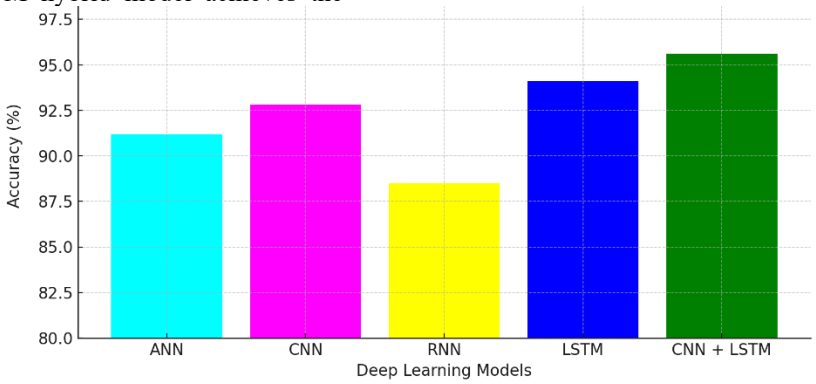


FIGURE 6: DEEP LEARNING MODEL PERFORMANCE IN HEART DISEASE PREDICTION

VII. CONCLUSIONS

Heart disease remains a significant global health concern, necessitating advanced predictive models for early diagnosis and prevention. This study demonstrates that both Machine Learning (ML) and Deep Learning (DL) techniques play a vital role in enhancing heart disease prediction through clinical datasets, medical imaging, and physiological signals. Traditional ML models, including Random Forest, Support Vector Machines (SVM), and XGBoost, are efficient for structured data analysis, while DL models such as CNN and RNN provide superior performance in analyzing ECG signals and medical images. The Hybrid CNN + LSTM model outperforms individual models by combining spatial feature extraction with sequential data processing, leading to higher predictive accuracy.

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